Evaluation Report: Carefull Alert System.

1. Classification Metrics The system was evaluated against 291 ground-truth transactions using a two-stage pipeline (Heuristic filtering + LLM validation).

Model A: **OpenRouter (GPT-4o-mini)**

- Precision: 1.00
- Recall: 0.72
- F1-Score: 0.84

Model B: **Groq (Llama-3-70b)**

- Precision: 1.00
- Recall: 0.75
- F1-Score: 0.86

2. Confusion Matrix (Groq / Llama-3)

- True Negatives (Normal -> Normal): 259
- False Positives (Normal -> Fraud): 0
- False Negatives (Fraud -> Normal): 8
- True Positives (Fraud -> Fraud): 24

Error Distribution Analysis: The system achieved a 0% False Positive rate, meaning no benign transactions were incorrectly flagged. The errors consist entirely of False Negatives (8 missed alerts). These occurred because the initial deterministic heuristic thresholds were set conservatively (e.g., requiring 3 transactions for an escalating pattern, or >$2000 for money mules) to prevent flooding the LLM. The missed frauds were early-stage anomalies that mathematically blended with normal spending.

3. **Model Comparison**
   Metric | OpenRouter (GPT-4o-mini) | Groq (Llama-3-70b)
   Accuracy | 0.97 | 0.97
   Avg. Latency (per LLM call) | 2.55s | 2.83s
   Avg. Confidence Score | 83.8% | 85.7%
   Frauds Caught | 23 | 24
4. **Cost Projection (50,000 tx/month)**
   Because the Stage 1 Heuristic Filter drops ~90% of normal transactions, only ~5,000 transactions require LLM processing per month.

- **OpenRouter (GPT-4o-mini):** ~$0.00015 per transaction.
  - Projected Monthly Cost: $0.75 USD
- **Groq (Commercial rate estimate):** ~$0.00060 per transaction.
  - Projected Monthly Cost: $3.00 USD